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### Air intake for an aircraft propulsion system engine assembly

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#### Abstract

An engine assembly for an aircraft propulsion system includes a gas turbine engine, a gearbox, and a first drive shaft. The gas turbine engine includes an engine output shaft, an air intake, and a compressor. The air intake includes an air intake body forming an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet. The air plenum extends circumferentially about the engine output shaft between and to a first circumferential end of the air intake and a second circumferential end of the air intake. The first circumferential end and the second circumferential end form a circumferential gap. The compressor is connected in fluid communication with the air intake. The gearbox includes a gear assembly operably connected to the engine output shaft. The first drive shaft is operably connected to the gear assembly. The first drive shaft extends through the circumferential gap.

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**Inventors:** Desjardins; Michel (Saint-Hubert, CA), Landry-Drolet; Guillaume (Boucherville, CA)

**Applicant:** Pratt & Whitney Canada Corp. (Longueuil, CA)

**Family ID:** 92909489

**Assignee:** Pratt & Whitney Canada Corp. (Longueuil, CA)

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*Primary Examiner:* Chau; Alain

*Attorney, Agent or Firm:* Getz Balich LLC

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## Background/Summary

### TECHNICAL FIELD

(1) This disclosure relates generally to engine assemblies configured for aircraft propulsion systems

and, more particularly, to an air intake for an engine assembly.

## BACKGROUND OF THE ART

(2) A propulsion system for an aircraft may include an air intake configured to direct ambient air into a gas turbine engine of the propulsion system. Various air intake configurations are known in the art for directing air into a propulsion system gas turbine engine. While these known air intakes have various advantages, there is still room in the art for improvement.

## SUMMARY

(3) It should be understood that any or all of the features or embodiments described herein can be used or combined in any combination with each and every other feature or embodiment described herein unless expressly noted otherwise.

(4) According to an aspect of the present disclosure, an engine assembly for an aircraft propulsion system includes a gas turbine engine, a gearbox, and a first drive shaft. The gas turbine engine includes an engine output shaft, and air intake, and a compressor. The engine output shaft is configured for rotation about a first rotational axis. The air intake includes an air intake body forming an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet in fluid communication. The air plenum extends circumferentially about the engine output shaft between and to a first circumferential end of the air intake and a second circumferential end of the air intake. The first circumferential end and the second circumferential end form a circumferential gap outside of the air intake and circumferentially between the first circumferential end and the second circumferential end. The compressor is connected in fluid communication with the air intake at the air outlet. The gearbox includes a gear assembly operably connected to the engine output shaft. The first drive shaft is operably connected to the gear assembly. The first drive shaft is rotatable about a second rotational axis. The second rotational axis is radially offset from the first rotational axis. The first drive shaft extends along the second rotational axis through the circumferential gap.

(5) In any of the aspects or embodiments described above and herein, at least a portion of the first drive shaft may be disposed radially inside the air intake body.

(6) In any of the aspects or embodiments described above and herein, the air intake may further include a screen. The screen may extend circumferentially within the air plenum from the first circumferential end to the second circumferential end.

(7) In any of the aspects or embodiments described above and herein, at least a portion of the first drive shaft may be disposed radially inside the screen.

(8) In any of the aspects or embodiments described above and herein, the engine assembly may further include a second drive shaft operably connected to the gear assembly. The second drive shaft may be rotatable about a third rotational axis. The third rotational axis may be radially offset from the first rotational axis.

(9) In any of the aspects or embodiments described above and herein, the third rotational axis and the second rotational axis may be colinear.

(10) In any of the aspects or embodiments described above and herein, the air intake body may form the air inlet at a first circumferential position of the air intake.

(11) In any of the aspects or embodiments described above and herein, the first circumferential end and the second circumferential end may form the circumferential gap at a second circumferential position circumferentially opposite the first circumferential position.

(12) In any of the aspects or embodiments described above and herein, the air intake may include an end strut forming the first circumferential end and the second circumferential end.

(13) In any of the aspects or embodiments described above and herein, the air intake body may include an outer radial wall portion. The outer radial wall portion may be radially offset from the air outlet.

(14) In any of the aspects or embodiments described above and herein, a radial distance between the outer radial wall portion and the air outlet may decrease in a circumferential direction from the

air inlet to the first circumferential end and from the air inlet to the second circumferential end.

(15) In any of the aspects or embodiments described above and herein, the air intake may further include at least one guide strut disposed within the air plenum.

(16) According to another aspect of the present disclosure, an air intake for an aircraft propulsion system engine assembly includes an air intake body. The air intake body extends circumferentially about an axis. The air intake body forms an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet in fluid communication. The air plenum extends circumferentially about the axis between and to a first circumferential end of the air intake and a second circumferential end of the air intake. The air intake body includes an air inlet body portion, a first wall portion, a second wall portion, and an outer radial wall portion. The air inlet body portion forms the air inlet at a first circumferential position of the air intake. The first wall portion and the second wall portion extend circumferentially from the air inlet body portion to the first circumferential end. The first wall portion and the second wall portion extend circumferentially from the air inlet body portion to the second circumferential end. The first wall portion and the second wall portion form the air plenum and the air outlet between the first wall portion and the second wall portion. The outer radial wall portion extends along the first wall portion and the second wall portion and further forms the air plenum. The outer radial wall portion extends circumferentially from the air inlet body portion to the first circumferential end. The outer radial wall portion extends circumferentially from the air inlet body portion to the second circumferential end.

(17) In any of the aspects or embodiments described above and herein, the first circumferential end and the second circumferential end may form a circumferential gap outside of the air intake and circumferentially between the first circumferential end and the second circumferential end.

(18) In any of the aspects or embodiments described above and herein, the first circumferential end and the second circumferential end may form the circumferential gap at a second circumferential position circumferentially opposite the first circumferential position.

(19) In any of the aspects or embodiments described above and herein, the air intake may include an end strut extending between and connecting the first wall portion, the second wall portion, and the outer radial wall portion. The end strut may form the first circumferential end and the second circumferential end.

(20) According to another aspect of the present disclosure, an engine assembly for an aircraft propulsion system includes a gas turbine engine and a drive shaft. The gas turbine engine includes an air intake and a compressor. The air intake includes an air intake body extending circumferentially about an axis. The air intake body forms an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet in fluid communication. The air intake body forms the air inlet at a first circumferential position of the air intake. The air plenum extends circumferentially about the axis between and to a first circumferential end of the air intake and a second circumferential end of the air intake. The first circumferential end and the second circumferential end form a circumferential gap outside of the air intake and circumferentially between the first circumferential end and the second circumferential end. The first circumferential end and the second circumferential end form the circumferential gap at a second circumferential position of the air intake circumferentially opposite the first circumferential position. The compressor is connected in fluid communication with the air intake at the air outlet. The drive shaft is rotatable about a rotational axis. The rotational axis is radially offset from the axis. The drive shaft extends along the rotational axis through the circumferential gap.

(21) In any of the aspects or embodiments described above and herein, at least a portion of the drive shaft may be disposed radially inside the air intake body.

(22) In any of the aspects or embodiments described above and herein, the air intake may further include a screen. The screen may extend circumferentially within the air plenum from the first circumferential end to the second circumferential end.

(23) In any of the aspects or embodiments described above and herein, at least a portion of the drive shaft may be disposed radially inside the screen.

(24) The present disclosure, and all its aspects, embodiments and advantages associated therewith will become more readily apparent in view of the detailed description provided below, including the accompanying drawings.

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## Description

### DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 illustrates a perspective view of an aircraft including a propulsion system, in accordance with one or more embodiments of the present disclosure.

(2) FIG. 2 schematically illustrates a side, cutaway view of an engine assembly for an aircraft propulsion system, in accordance with one or more embodiments of the present disclosure.

(3) FIG. 3 schematically illustrates a side, cutaway view of a gas turbine engine for an aircraft propulsion system, in accordance with one or more embodiments of the present disclosure.

(4) FIG. 4 schematically illustrates a front, cutaway view of an air intake for the engine assembly of FIG. 2, in accordance with one or more embodiments of the present disclosure.

(5) FIG. 5 schematically illustrates a side, cutaway view of the air intake of FIG. 4, in accordance with one or more embodiments of the present disclosure.

(6) FIG. 6 schematically illustrates a front, cutaway view of another air intake for the engine assembly of FIG. 2, in accordance with one or more embodiments of the present disclosure.

(7) FIGS. 7A-B illustrate perspective, cutaway views of a portion of an air intake including a guide strut, in accordance with one or more embodiments of the present disclosure.

### DETAILED DESCRIPTION

(8) FIG. 1 illustrates a perspective view of an aircraft **1000** including a propulsion system **10**. The aircraft **1000** of FIG. 1 is configured as a rotary-wing aircraft (e.g., a helicopter). The present disclosure, however, is also applicable to other configurations of aircraft such as, but not limited to, a fixed-wing aircraft (e.g., an airplane), a tilt-rotor aircraft, a tilt-wing aircraft, or another aerial vehicle. The propulsion system **10** of FIG. 1 is configured to drive rotation of a main rotor **1002** and a tail rotor **1004** of the aircraft **1000** to facilitate propulsion for the aircraft **1000**. The aircraft **1000** of FIG. 1 further includes a housing **1006** (e.g., a nacelle, a fuselage, etc.) enclosing and forming a protective outer covering for the propulsion system **10**.

(9) FIG. 2 schematically illustrates a cutaway, side view of the propulsion system **10**. The propulsion system **10** of FIG. 2 includes an engine assembly **20**. The engine assembly **20** includes a gas turbine engine **22**, an engine transmission gearbox **24**, a main rotor drive shaft **26**, a main rotor gearbox **28**, a tail rotor drive shaft **30**, a tail rotor gearbox **32**, and an air intake **34**.

(10) FIG. 3 schematically illustrates a cutaway, side view of the gas turbine engine **22**. The gas turbine engine **22** of FIG. 3 includes a compressor section **36**, a combustor section **38**, a turbine section **40**, and an engine static structure **42**. The compressor section **36** includes a compressor **44**. The combustor section **38** includes a combustor **46** (e.g., an annular combustor). The turbine section **40** includes a high-pressure turbine **48** and a power turbine **50**. The gas turbine engine **22** sections **36**, **38**, **40** of FIG. 3 are sequentially arranged along an axial centerline **52** (e.g., a rotational axis) of the gas turbine engine **22**. The engine static structure **42** may include, for example, one or more engine cases for the gas turbine engine **22**. The engine static structure **42** may additionally include cowlings, bearing assemblies, or other structural components of the gas turbine engine **22**. The engine static structure **42** houses and structurally supports the engine sections **36**, **38**, **40**.

(11) Components of the engine sections **26**, **28**, **30** form a first rotational assembly **54** (e.g., a high-pressure spool) and a second rotational assembly **56** (e.g., a power spool). The first rotational

assembly **54** and the second rotational assembly **56** are mounted for rotation about the axial centerline **52** relative to the engine static structure **42**.

(12) The first rotational assembly **54** includes a first shaft **58**, a bladed compressor rotor **60** for the compressor **44**, and a bladed first turbine rotor **62** for the high-pressure turbine **48**. The first shaft **58** interconnects the bladed compressor rotor **60** and the bladed first turbine rotor **62**.

(13) The second rotational assembly **56** includes a second shaft **64** and a bladed second turbine rotor **66** for the power turbine **50**. The second shaft **64** is a power shaft (e.g., an engine output shaft) of the gas turbine engine **22**. The second shaft **64** interconnects the bladed second turbine rotor **66** with the engine transmission gearbox **24**. The first shaft **58** and the second shaft **64** are concentric and configured to rotate about the axial centerline **52**. The present disclosure, however, is not limited to concentric configurations of the first shaft **58** and the second shaft **64**.

(14) During operation of the gas turbine engine **22**, ambient air is directed into the compressor section **36** by the air intake **34**. The air is compressed in the compressor **44** by the bladed compressor rotor **60** and directed into a combustion chamber of the combustor **46**. Fuel is injected into the combustion chamber and mixed with the compressed air to provide a fuel-air mixture. This fuel-air mixture is ignited and combustion products thereof flow through and sequentially cause the bladed first turbine rotor **62** and the bladed second turbine rotor **66** to rotate. The rotation of the bladed first turbine rotor **62** and the bladed second turbine rotor **66** respectively drives rotation of the first rotational assembly **54** and the second rotational assembly **56**.

(15) Referring again to FIG. **2**, the engine transmission gearbox **24** of FIG. **2** includes a gear assembly **68**. The gear assembly **68** is operably connected to (e.g., coupled with) the second shaft **64**, the main rotor drive shaft **26**, and the tail rotor drive shaft **30**. The gas turbine engine **22** is configured to drive rotation of the main rotor drive shaft **26** and the tail rotor drive shaft **30** with the second shaft **64** through the gear assembly **68**. The gear assembly **68** operably connects the second shaft **64**, the main rotor drive shaft **26**, and the tail rotor drive shaft **30**. The gear assembly **68** may be configured, for example, an offset gear assembly, an epicyclic gear assembly (e.g., a planetary gear assembly), or another suitable gear assembly for operably connecting the second shaft **64**, the main rotor drive shaft **26**, and the tail rotor drive shaft **30**. The present disclosure, however, is not limited to any particular gear configuration for the gear assembly **68**.

(16) The main rotor drive shaft **26** operably interconnects (e.g., couples) the engine transmission gearbox **24** (e.g., the gear assembly **68**) with the main rotor gearbox **28**. The main rotor drive shaft **26** may extend in an axially forward direction from the engine transmission gearbox **24** to the main rotor gearbox **28** (e.g., relative to the axial centerline **52**). The main rotor drive shaft **26** extends along a rotational axis **70** between and to a first axial end **72** of the main rotor drive shaft **26** and a second axial end **74** of the main rotor drive shaft **26**. The main rotor drive shaft **26** is rotatable about the rotational axis **70**. The rotational axis **70** may extend parallel to or substantially parallel to the axial centerline **52**. The present disclosure, however, is not limited to any particular orientation of the rotational axis **70** relative to the axial centerline **52**. The main rotor drive shaft **26** and its rotational axis **70** may be radially offset from the second shaft **64** (e.g., with respect to the axial centerline **52**). The main rotor drive shaft **26** may be disposed vertically below the second shaft **64** as shown, for example, in FIG. **2**. The present disclosure, however, is not limited to any particular vertical position of the main rotor drive shaft **26** relative to the second shaft **64**. As used herein, the term “vertical” and associated orientational language such as “vertically above,” “vertically below,” “up,” and “down” should be understood with respect to a normal attitude of the propulsion system **10**. The main rotor drive shaft **26** may be coupled with the gear assembly **68** at (e.g., on, adjacent, or proximate) the first axial end **72**. The main rotor drive shaft **26** may be coupled with a gear assembly **76** of the main rotor gearbox **28** at (e.g., on, adjacent, or proximate) the second axial end **74**. The gear assembly **76** is further operably connected to (e.g., coupled with) the main rotor **1002** (see FIG. **1**) to drive rotation of the main rotor **1002** in response to rotation of the main rotor drive shaft **26**.

(17) The tail rotor drive shaft **30** operably interconnects (e.g., couples) the engine transmission gearbox **24** (e.g., the gear assembly **68**) with the tail rotor gearbox **32**. The tail rotor drive shaft **30** may extend in an axially aft direction from the engine transmission gearbox **24** to the tail rotor gearbox **32** (e.g., relative to the axial centerline **52**). As will be discussed in further detail, the tail rotor drive shaft **30** is disposed axially and radially coincident with the air intake **34**, relative to the axial centerline **52**. The tail rotor drive shaft **30** extends along a rotational axis **78** between and to a first axial end **80** of the tail rotor drive shaft **30** and a second axial end **82** of the tail rotor drive shaft **30**. The tail rotor drive shaft **30** is rotatable about the rotational axis **78**. The rotational axis **78** may extend parallel to or substantially parallel to the axial centerline **52**. The rotational axis **78** may additionally extend parallel to, substantially parallel to, or colinear with the rotational axis **70**. The present disclosure, however, is not limited to any particular orientation of the rotational axis **78** relative to the axial centerline **52** or the rotational axis **70**. The tail rotor drive shaft **30** and its rotational axis **78** are radially offset from the second shaft **64** (e.g., with respect to the axial centerline **52**). The tail rotor drive shaft **30** may be disposed vertically below the second shaft **64** as shown, for example, in FIG. 2. The present disclosure, however, is not limited to any particular vertical position of the tail rotor drive shaft **30** relative to the second shaft **64**. The tail rotor drive shaft **30** may be coupled with the gear assembly **68** at (e.g., on, adjacent, or proximate) the first axial end **80**. The first axial end **80** may be disposed axially forward of the air intake **34**. The tail rotor drive shaft **30** may be coupled with a gear assembly **84** of the tail rotor gearbox **32** at (e.g., on, adjacent, or proximate) the second axial end **82**. The second axial end **82** may be disposed axially aft of the air intake **34**. The gear assembly **84** is further operably connected to (e.g., coupled with) the tail rotor **1004** (see FIG. 1) to drive rotation of the tail rotor **1004** in response to rotation of the tail rotor drive shaft **30**.

(18) The main rotor drive shaft **26** and the tail rotor drive shaft **30** may be formed by a single drive shaft extending through the engine transmission gearbox **24** along the rotational axis **70** and the rotational axis **78** (e.g., colinear rotational axes). Accordingly, the single drive shaft forming the main rotor drive shaft **26** and the tail rotor drive shaft **30** may extend between and to the second axial end **74** and the second axial end **82**. The single drive shaft may be coupled with the gear assembly **76** at (e.g., on, adjacent, or proximate) the second axial end **74**. The single drive shaft may be coupled with the gear assembly **84** at (e.g., on, adjacent, or proximate) the second axial end **82**. The single drive shaft may be coupled with the gear assembly **68** at (e.g., on, adjacent, or proximate) an axially-intermediate portion of the single drive shaft between the second axial end **74** and the second axial end **82**.

(19) FIG. 4 schematically illustrates a front, cutaway view of the air intake **34**, the second shaft **64**, and the tail rotor drive shaft **30**. FIG. 5 schematically illustrates a side, cutaway view of the air intake **34**, the second shaft **64**, the tail rotor drive shaft **30**, and portions of the compressor section **36**.

(20) The air intake **34** includes an air intake body **86**. The air intake **34** may additionally include a screen **102**. The air intake body **86** forms an air inlet **88**, an air outlet **90**, and an air plenum **92** of the air intake **34**. The air inlet **88** is disposed at (e.g., on, adjacent, or proximate) an exterior of the propulsion system **10** (see FIG. 2). Ambient air outside the propulsion system **10** enters the air intake **34** at the air inlet **88**. The air outlet **90** is disposed at (e.g., on, adjacent, or proximate) the compressor section **36** (e.g., an air inlet of the compressor **44**) (see FIG. 3). Air from the air intake **34** flows into the compressor **44** through the air outlet **90**. The air plenum **92** extends between and connects the air inlet **88** and the air outlet **90** in fluid communication. The air intake body **86** may be formed by a single (e.g., unitary) component. Alternatively, the air intake body **86** may include a plurality of components assembled together to form the air intake body **86**.

(21) The air intake body **86** of FIGS. 4 and 5 includes an air inlet body portion **94**, a first wall portion **96**, a second wall portion **98**, and an outer radial wall portion **100**. The air inlet body portion **94** is disposed at a circumferential position of the air intake **34** (e.g., relative to the axial

centerline 52). For example, the air inlet body portion 94 of FIGS. 4 and 5 may be understood to be disposed at a vertical top position (e.g., a 12 o'clock position) of the air intake 34. The air inlet body portion 94 surrounds and forms the air inlet 88 and directs ambient air through the air inlet 88 and into the air plenum 92. The first wall portion 96, the second wall portion 98, and the outer radial wall portion 100 surround and form the air plenum 92. The first wall portion 96, the second wall portion 98, and the outer radial wall portion 100 extend circumferentially from the air inlet body portion 94 to a first circumferential end 104 of air intake 34 and a second circumferential end 106 of the air intake 34, relative to the axial centerline 52. The first wall portion 96 and the second wall portion 98 may curve from a radial orientation toward an axial orientation in a direction from the air inlet 88 to the air outlet 90. For example, the first wall portion 96 may be disposed axially forward of the second wall portion 98 at (e.g., on, adjacent, or proximate) the air inlet 88 and the first wall portion 96 may be disposed radially inward of the second wall portion 98 at (e.g., on, adjacent, or proximate) the air outlet 90. The first wall portion 96 may converge toward the second wall portion 98 in a direction from the air inlet 88 to the air outlet 90, however, the present disclosure is not limited to this particular configuration of the air intake body 86. The outer radial wall portion 100 extends circumferentially along the first wall portion 96 and the second wall portion 98 forming an outer radial boundary of the air plenum 92.

(22) The first circumferential end 104 and the second circumferential end 106 enclose and form opposing circumferential ends of the air plenum 92. The first circumferential end 104 and the second circumferential end 106 form a circumferential gap 108 between the first circumferential end 104 and the second circumferential end 106. In other words, the circumferential gap 108 extends circumferentially between and to the first circumferential end 104 and the second circumferential end 106 outside of the air intake 34. The first circumferential end 104 and the second circumferential end 106 may form the circumferential gap 108 at a circumferential position of the air intake 34 (e.g., relative to the axial centerline 52) which is circumferentially opposite the air inlet body portion 94 and the air inlet 88. For example, the circumferential gap 108 of FIG. 4 may be understood to be disposed at a vertical bottom position (e.g., a 6 o'clock position) of the air intake 34.

(23) The air intake 34 may include an end strut 110 forming the first circumferential end 104 and the second circumferential end 106. The end strut 110 may also form the circumferential gap 108. The end strut 110 may extend between and connect the first wall portion 96, the second wall portion 98, and the outer radial wall portion 100 at (e.g., on, adjacent, or proximate) the first circumferential end 104 to enclose (e.g., seal) the air plenum 92 at the first circumferential end 104. Similarly, the end strut 110 may extend between and connect the first wall portion 96, the second wall portion 98, and the outer radial wall portion 100 at (e.g., on, adjacent, or proximate) the second circumferential end 106 to enclose (e.g., seal) the air plenum 92 at the second circumferential end 106. The end strut 110 may extend radially or substantially radially from the outer radial wall portion 100 toward the axial centerline 52. The present disclosure, however, is not limited to any particular orientation of the end strut 110 relative to the axial centerline 52. The first circumferential end 104 and the second circumferential end 106 may alternatively be formed by portions of the air intake body 86.

(24) The air intake 34 may include the screen 102 disposed within the air plenum 92. The screen 102 may extend circumferentially within the air plenum 92 between and to the first circumferential end 104 and the second circumferential end 106. For example, the screen 102 may extend circumferentially between and to the end strut 110 at (e.g., on, adjacent, or proximate) the first circumferential end 104 and the end strut 110 at (e.g., on, adjacent, or proximate) the second circumferential end 106. The screen 102 may extend axially within the air plenum 92 between and to the first wall portion 96 and the second wall portion 98. The screen 102 is disposed between the air inlet 88 and the air outlet 90 with respect to an air flow path through the air intake 34 and into the compressor section 36 (see FIG. 3). The screen 102 may be formed, for example, by a wire



mesh, a perforated plate, or other porous structure configured to prevent larger particulate entering the air inlet **88** from flowing to the air outlet **90**.

(25) Referring again to FIGS. **2**, **4**, and **5**, the tail rotor drive shaft **30** extends through the circumferential gap **108** between (e.g., axially between) the first axial end **80** and the second axial end **82**. The tail rotor drive shaft **30** is positioned within the circumferential gap **108** and circumferentially adjacent the first circumferential end **104** (e.g., the end strut **110**) and the second circumferential end **106** (e.g., the end strut **112**). For example, at least a portion of the tail rotor drive shaft **30** disposed in the circumferential gap **108** may be radially inside of the outer radial wall portion **100**. For further example, all of the tail rotor drive shaft **30** disposed in the circumferential gap **108** may be radially inside of the outer radial wall portion **100**. For further example, at least a portion of the tail rotor drive shaft **30** disposed in the circumferential gap **108** may be radially inside of the screen **102**.

(26) The circumferential gap **108** formed by the air intake **34** and the position of the tail rotor drive shaft **30** within the circumferential gap **108** facilitates a reduction in radial distance **D1** between the axial centerline **52** and the rotational axis **78** (e.g., the tail rotor drive shaft **30**), relative to a conventional air intake which extends fully, circumferentially about (e.g., completely around) an axial centerline and/or engine output shaft of a gas turbine engine. Accordingly, the present disclosure air intake facilitates a more radially-compact engine assembly **20**. The more radially-compact engine assembly **20** facilitates a reduction in size and/or material of the housing **1006** (see FIG. **1**) and other engine assembly **20** components (e.g., the air intake **34**, the engine transmission gearbox **24**, etc.), thereby reducing a total weight of the propulsion system **10** and facilitate an improved power-to-weight ratio of the propulsion system **10**. Reducing the size of the housing **1006** may further facilitate a reduction in aerodynamic drag for the propulsion system **10** and the aircraft **1000** (see FIG. **1**). For conventional air intakes in which the air plenum is fully annular, we have found that an air stagnation region may be formed at a circumferential position opposite the air inlet (e.g., at the vertical bottom or 6 o'clock position). The present disclosure air intake **34** may form the circumferential gap **108** at this position circumferentially opposite the air inlet **88** and, therefore, may have a negligible impact on air flow supplied to the compressor section **36** (see FIG. **3**).

(27) Referring to FIG. **6**, in some embodiments, the air intake **34** may be configured with the outer radial wall portion **100** radially offset relative to the air outlet **90**. For example, a radial distance **D2** between the outer radial wall portion **100** and the air outlet **90** may decrease in a circumferential direction from the air inlet **88** to the first circumferential end **104** and from the air inlet **88** to the second circumferential end **106**. This configuration of the air intake **34** of FIG. **6** may facilitate improved air flow distribution at the air outlet **90**. In particular, the air intake **34** of FIG. **6** may facilitate more circumferentially uniform air flow supplied to the compressor section **36** (see FIG. **3**) (e.g., the compressor entry face) with respect to air pressure, air flow rate, and air swirl.

(28) Referring to FIGS. **7A-B**, in some embodiments, the air intake **34** may include one or more guide struts **114** disposed within the air plenum **92**. The guide struts **114** may be formed by or otherwise connected to the air intake body **86**. The guide struts **114** may be circumferentially distributed throughout the air plenum **92**. FIGS. **7A-B** illustrate cutaway views of one of the guide struts **114** projecting outward from the second wall portion **98**. Each of the guide struts **114** may extend between and connect the first wall portion **96** and the second wall portion **98** (see FIG. **5**). Some or all of the guide struts **114** may extend through the first wall portion **96** and connect to the gearbox **24** (see FIG. **2**). For example, the guide struts **114** may be structural struts connecting the gearbox **24** and the gas turbine engine **22** (e.g., the engine static structure **42**; see FIG. **3**) through the air intake **34**. As shown in FIGS. **7A-B**, the guide struts **114** may be oriented to induce swirl or otherwise direct air flow in the air plenum **92** in a clockwise or counterclockwise direction to control air flow and air distribution within the air plenum **92**.

(29) While the principles of the disclosure have been described above in connection with specific

apparatuses and methods, it is to be clearly understood that this description is made only by way of example and not as limitation on the scope of the disclosure. Specific details are given in the above description to provide a thorough understanding of the embodiments. However, it is understood that the embodiments may be practiced without these specific details.

(30) It is noted that the embodiments may be described as a process which is depicted as a flowchart, a flow diagram, a block diagram, etc. Although any one of these structures may describe the operations as a sequential process, many of the operations can be performed in parallel or concurrently. In addition, the order of the operations may be rearranged. A process may correspond to a method, a function, a procedure, a subroutine, a subprogram, etc.

(31) The singular forms “a,” “an,” and “the” refer to one or more than one, unless the context clearly dictates otherwise. For example, the term “comprising a specimen” includes single or plural specimens and is considered equivalent to the phrase “comprising at least one specimen.” The term “or” refers to a single element of stated alternative elements or a combination of two or more elements unless the context clearly indicates otherwise. As used herein, “comprises” means “includes.” Thus, “comprising A or B,” means “including A or B, or A and B,” without excluding additional elements.

(32) It is noted that various connections are set forth between elements in the present description and drawings (the contents of which are included in this disclosure by way of reference). It is noted that these connections are general and, unless specified otherwise, may be direct or indirect and that this specification is not intended to be limiting in this respect. Any reference to attached, fixed, connected, or the like may include permanent, removable, temporary, partial, full and/or any other possible attachment option.

(33) No element, component, or method step in the present disclosure is intended to be dedicated to the public regardless of whether the element, component, or method step is explicitly recited in the claims. No claim element herein is to be construed under the provisions of 35 U.S.C. 112(f) unless the element is expressly recited using the phrase “means for.” As used herein, the terms “comprise”, “comprising”, or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus.

(34) While various inventive aspects, concepts and features of the disclosures may be described and illustrated herein as embodied in combination in the exemplary embodiments, these various aspects, concepts, and features may be used in many alternative embodiments, either individually or in various combinations and sub-combinations thereof. Unless expressly excluded herein all such combinations and sub-combinations are intended to be within the scope of the present application. Still further, while various alternative embodiments as to the various aspects, concepts, and features of the disclosures—such as alternative materials, structures, configurations, methods, devices, and components, and so on—may be described herein, such descriptions are not intended to be a complete or exhaustive list of available alternative embodiments, whether presently known or later developed. Those skilled in the art may readily adopt one or more of the inventive aspects, concepts, or features into additional embodiments and uses within the scope of the present application even if such embodiments are not expressly disclosed herein. For example, in the exemplary embodiments described above within the Detailed Description portion of the present specification, elements may be described as individual units and shown as independent of one another to facilitate the description. In alternative embodiments, such elements may be configured as combined elements.

## Claims

1. An engine assembly for an aircraft propulsion system, the engine assembly comprising: a gas turbine engine including: an engine output shaft configured for rotation about a first rotational axis, an air intake including an air intake body forming an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet in fluid communication, the air plenum extends circumferentially about the engine output shaft between and to a first circumferential end of the air intake and a second circumferential end of the air intake, the first circumferential end and the second circumferential end enclosing the air plenum opposite the air inlet relative to the first rotational axis, and the first circumferential end and the second circumferential end form a circumferential gap outside of the air intake and circumferentially between the first circumferential end and the second circumferential end, the circumferential gap located opposite the air inlet relative to the first rotational axis, and a compressor connected in fluid communication with the air intake at the air outlet; a gearbox including a gear assembly operably connected to the engine output shaft; and a first drive shaft operably connected to the gear assembly, the first drive shaft is rotatable about a second rotational axis, the second rotational axis is radially offset from the first rotational axis, and the first drive shaft extends along the second rotational axis through the circumferential gap.
2. The engine assembly of claim 1, wherein at least a portion of the first drive shaft is disposed radially inside the air intake body.
3. The engine assembly of claim 1, wherein the air intake further includes a screen, and the screen extends circumferentially within the air plenum from the first circumferential end to the second circumferential end.
4. The engine assembly of claim 3, wherein at least a portion of the first drive shaft is disposed radially inside the screen.
5. The engine assembly of claim 1, further comprising a second drive shaft operably connected to the gear assembly, the second drive shaft is rotatable about a third rotational axis, and the third rotational axis is radially offset from the first rotational axis.
6. The engine assembly of claim 5, wherein the third rotational axis and the second rotational axis are colinear.
7. The engine assembly of claim 1, wherein the air intake body forms the air inlet at a first circumferential position of the air intake.
8. The engine assembly of claim 1, wherein the air intake includes an end strut forming the first circumferential end and the second circumferential end.
9. The engine assembly of claim 1, wherein the air intake body includes an outer radial wall portion, and the outer radial wall portion is radially offset from the air outlet.
10. The engine assembly of claim 9, wherein a radial distance between the outer radial wall portion and the air outlet decreases in a circumferential direction from: the air inlet to the first circumferential end; and the air inlet to the second circumferential end.
11. The engine assembly of claim 1, wherein the air intake further includes at least one guide strut disposed within the air plenum.
12. The engine assembly of claim 1, wherein the engine assembly is enclosed by a housing.
13. The engine assembly of claim 1, wherein the air inlet is disposed radially outward of the air intake.
14. An engine assembly for an aircraft propulsion system, the engine assembly comprising: a gas turbine engine including: an air intake including an air intake body extending circumferentially about an axis, the air intake body forms an air inlet, an air outlet, and an air plenum extending between and connecting the air inlet and the air outlet in fluid communication, the air intake body forms the air inlet at a first circumferential position of the air intake, the air plenum extends circumferentially about the axis between and to a first circumferential end of the air intake and a second circumferential end of the air intake, the first circumferential end and the second

circumferential end form a circumferential gap outside of the air intake and circumferentially between the first circumferential end and the second circumferential end, and the first circumferential end, the second circumferential end, and the circumferential gap disposed at a second circumferential position of the air intake circumferentially opposite the first circumferential position; a compressor connected in fluid communication with the air intake at the air outlet; and a drive shaft, the drive shaft is rotatable about a rotational axis, the rotational axis is radially offset from the axis, and the drive shaft extends along the rotational axis through the circumferential gap.

15. The engine assembly of claim 14, wherein at least a portion of the drive shaft is disposed radially inside the air intake body.

16. The engine assembly of claim 14, wherein the air intake further includes a screen, and the screen extends circumferentially within the air plenum from the first circumferential end to the second circumferential end.

17. The engine assembly of claim 16, wherein at least a portion of the drive shaft is disposed radially inside the screen.

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